

MA.6.AR.3.3

Benchmark

MA.6.AR.3.3 **Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios.**

Example: The table below expresses the relationship between the number of ounces of yellow and blue paints used to create a new color. Determine the ratios and complete the table.

Yellow (part)	1.5	3		9
Blue (part)	2	4		
New color (whole)			12	21

Benchmark Clarifications:

Clarification 1: Instruction includes using two-column tables (e.g., a relationship between two variables) and three-column tables (e.g., part-to-part-to-whole relationship) to generate conversion charts and mixture charts.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.2.1, MA.6.NSO.2.3
- MA.6.NSO.3.1
- MA.6.AR.2.2, MA.6.AR.2.3

Terms from the K-12 Glossary

- Rate
- Unit Rate

Vertical Alignment

Previous Benchmarks

- MA.5.FR.1.1
- MA.5.FR.2.1, MA.5.FR.2.2

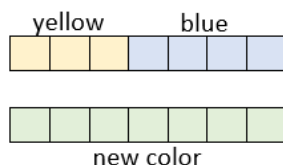
Next Benchmarks

- MA.7.AR.3.2
- MA.7.AR.4.1, MA.7.AR.4.2, MA.7.AR.4.3, MA.7.AR.4.4

Purpose and Instructional Strategies

In elementary grades, students developed understanding of fraction concepts and operations, often in the context of the relationship that relates parts to wholes. In grade 6, students use tables to display parts to parts and parts to whole relationship, as preparation for understanding a constant ratio relationship between two quantities. In grade 7, students will achieve a more formal understanding of the constant of proportionality and proportional relationships, which can be represented by tables, equations and graphs.

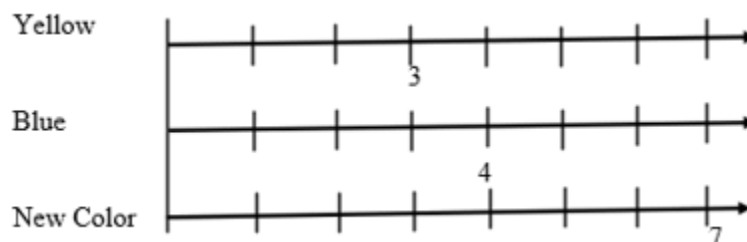
- Instruction includes discussion about how the patterns that students see in the tables can help them reason through various situations (*MTR.4.1, MTR.5.1*).
- Instruction includes the use of acting out scenarios, bar models, tables and number lines in order to see relationships and relevance to real-world application (*MTR.2.1*).
 - Bar Model



- Tables

Yellow (part)	3	
Blue (part)	4	
New color (whole)	7	

- Number Lines



Common Misconceptions or Errors

- Students may be looking for arithmetic or counting patterns within the table and not seeing multiplicative patterns between the ratio values.

Strategies to Support Tiered Instruction

- Instruction includes the use of whole number parts to allow for students to explore the various relationships in the tables before introducing positive rational values. Discussions include additive and multiplicative patterns amongst the rows and the columns of the table.
- Instruction includes the use of two different counters to represent the provided ratios with emphasis on the different colors representing the parts and the total number of counters representing the whole. Teacher allows for opportunities for the student to explore equivalent ratios by adding addition sets of counters or by dividing the existing counters into equal groups and recording the equivalent ratios in a two- or three-column table.
 - For example, students can use the counters to help complete the table to express the relationship between the number of students and the number of adults on a recent field trip.



Students (Red Counters)	3	6		
Adults (Yellow Counters)	1		3	
Total People (Total Counters)				16

Instructional Tasks

Instructional Task 1 (MTR.4.1, MTR.7.1)

Jeremy is making punch for a party. The relationship between the cups of syrup and the cups of water he used to create the punch mixture is shown on the table below.

Part A. Complete the table. Explain the process you used to complete the table.

syrup (cups)	water (cups)	punch (cups)
$1\frac{3}{4}$	7	
3	12	
	15	$18\frac{3}{4}$
		20

Part B. If Jeremy needs 30 cups of punch, how many cups of syrup are needed? How many cups of water are needed?

Instructional Task 2 (MTR.3.1)

Given the table below, determine the common ratio between the x - and y -values.

x	y
2	8
5	20
7	

Part A. Using a common ratio, complete the table when x is 7.

Part B. Using a common ratio, determine an x and corresponding y value.

Instructional Items

Instructional Item 1

Shambria is making pies and part of the recipe she uses is described in the table below.

Number of Pies	Cups of Sugar	Pounds of Apples
1	0.5	2.25
5		

Using this recipe, complete the table to determine the amount of sugar and apples needed to bake 5 pies.

Instructional Item 2

The table below describes the same rate at which Drew hiked along the Appalachian Trail each day. Complete the table and determine how many miles he could hike in 2 hours.

time (hours)	5		2
distance (miles)	20	12	

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*